www.csvtuonline.com

Printed Pages – 5 Roll No. .....

328351(14)

B. E. (Third Semester) Examination, April-May 2016

(New Scheme)

(ET&T Branch)

MATHEMATICS-III

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all questions. Part (a) of each question is compulsory and carries 2 marks. Solve any two parts from (b), (c) and (d) carrying 7 marks each.

1. (a) State shifting properties of Laplace transform.

(b) Find the Laplace transform of :

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(i) $f(t)=\begin{cases} t^2 & , \quad 0<t<2 \\ t-1 & , \quad 2<t<3 \\ 7 & , \quad t>3 \end{cases}$

(ii) $t\cos at$

(c) State and prove convolution theorem.

(d) Solve $\left(D^3-3D^2+3D-1\right)y=t^2e^t$ given that

$$y(0)=1, \quad y'(0)=0, \quad y''(0)=-2$$

2. (a) Define analytic function.

(b) If $f(z)$ is an analytic function of z, prove that :

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)\left|Rf(z)\right|^2=2\left|f'(z)\right|^2$$

(c) Find the Laurent's series expansion of

$\dfrac{z^2-6z-1}{(z-1)(z-3)(z+2)}$ in the region $3<|z+2|<5$

www.csvtuonline.com

www.csvtuonline.com

(d) Apply the calculus of residues to prove that :

$$\int_0^{2\pi} \frac{d\theta}{1-2p\sin\theta+p^2} = \frac{2\pi}{1-p^2} \quad (0<p<1)$$

3. (a) Show that the coefficient of correlation is the geometric mean of the coefficient of regression.

(b) Find the correlation coefficient between x and y from the given data :

x :	78	89	97	69	59	79	68	57
y :	125	137	156	112	107	138	123	108

(c) In a partially destroyed laboratory record, only the lines of regression of y on x and x on y are available as $4x-5y+3z=0$ and $20x-9y=107$ respectively. Calculate $\bar{x}$, $\bar{y}$ and the coefficient of co-relation between x and y.

(d) Prove that arithmetic mean of the coefficients of regression is greater than the coefficient of correlation.

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

4. (a) Write the Bessel's differential equation.

(b) Solve in series the equation :

$$9x(1-x)\frac{d^2y}{dx^2} - 12\frac{dy}{dx} + 4y = 0$$

(c) Prove that :

$$\frac{d}{dx}\left(J_n^2(x)+J_{n+1}^2(x)\right) = 2\left(\frac{n}{x}J_n^2(x) - \frac{n+1}{x}J_{n+1}^2(x)\right)$$

(d) Prove that the Legendre polynomials in algebraic form satisfy the orthogonality relations,

$$\int_{-1}^{1} P_m(x)P_n(x)\,dx = \begin{cases} 0 & , \text{ if } m \neq n \\ \dfrac{2}{2n+1} & , \text{ if } m = n \end{cases}$$

5. (a) Form the partial differential equation from

$$z = f(x+at) + g(x-at)$$

www.csvtuonline.com

www.csvtuonline.com

(b) Solve :

$$\left(x^2 - yz\right)p + \left(y^2 - zx\right)q = z^2 - xy$$

(c) Solve the following equation by the method of separation of variables :

$$3\frac{\partial u}{\partial x} + 2\frac{\partial u}{\partial y} = 0, \quad u(x, 0) = 4e^{-x}$$

(d) Neglecting R and G, find the e.m.f. $v(x, t)$ in a line of length l, t seconds after the ends were suddenly grounded, given that :

$$i(x, 0) = i_0 \qquad \text{and}$$

$$v(x, 0) = e_1 \sin\frac{\pi x}{l} + e_3 \sin\frac{5\pi x}{l}$$

Gujarat College of Engineering
ACC No....
BHUJ

www.csvtuonline.com

www.csvtuonline.com